

Machine Learning Model Based on REST API for Predicting Tenders Winner

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Abstract—Machine Learning is a tool that can provide the ability to make good predictions using experience data. The problem found from the results of research observations is that the marketing department reports that PT Rekayasa Industri is currently experiencing a decline in revenue that has the potential to harm the company, one of the factors is lack of understanding and deciding whether the company should participate in project tenders or not with aspects namely price, partners, and competitors. The purpose of this study is to utilize machine learning in developing a back-end system to help the marketing department to determine the probability of winning the upcoming tender so that it can be integrated with internally owned applications. The purpose of this research is to utilize machine learning to determine the probability of winning the upcoming tender by developing a back-end system using 3 (three) classification algorithms, namely Logistic Regression, Decision Tree, and Random Forest with REST API technology. The results of this study indicate that Random Forest is the appropriate model in this case, namely predicting victory, being able to study experience data well, and of course being a model that can help minimize the analysis process carried out by the marketing department. This is evidenced by the results of the tests carried out, namely the accuracy value of 96.15%, f1 score of 97.14% and the AUC value of 0.97 (very good classification). Followed by Decision Tree with an accuracy value of 84.62%, an f1 score of 89.47% and an AUC value of only 0.78 (classification is quite good). Meanwhile, Logistic Regression resulted in less than optimal data learning performance with an accuracy value of 80.77%, an f1 score of 87.18% and the AUC value which was quite good in the classification task, which was only 0.72.

Keywords—Python, Website, Classification, Logistic Regression, Decision Tree, Random Forest, Prediction, Algorithm, REST API

I. INTRODUCTION

Project cooperation activities are usually carried out by project implementing companies with several vendors, this activity called tender[1], which refers to the process of submitting or bidding in terms of price, wholesale, or providing all goods to support the project process later. In the public procurement literature [2], "tenders or bidding", whether electronic or not, involves not only interaction with suppliers but also the actual selection of winners and therefore requires the use of "market mechanisms".

PT Rekayasa Industri [3] or commonly abbreviated as Rekind, is a subsidiary of Pupuk Indonesia which does business in the EPC (Engineering, Procurement and

Construction) sector focusing on the development of national strategic projects such as oil refineries, petrochemicals and mining.

Based on Rekind's 2019 Annual Report [4], profit for the year amounted to IDR 184.32 billion, or 125.22% of the RKAP target of IDR 147.20 billion and only reached 91.8% of the 2018 realization of IDR 200.78 billion. The decrease in net profit achievement occurred due to a loss in exchange rates in 2019 and the project performance in terms of cost or the Cost Performance Index (CPI) reached 1.10, meaning that the average project is implemented at a lower cost compared to the contract value. However, the CPI index of 1.10 decreased compared to 2018 which reached 1.12.

It is not uncommon to encounter potential losses for the company before participating in a project tender [5], this happened because the projects held require quite expensive financing and the dynamic conditions of competitors and partners, this will certainly affect the winning factor in project tender [6] that the company will participate in.

Therefore, the company needs to dig deeper information regarding what information can be known from the company's experience in participating in past tender projects as material to find out the potential win in the next project tender to avoid losses in the form of increased spending or overspending cost by the company.

As mentioned in the explanation above, the use of algorithms in various fields is currently very helpful in the decision-making process. The probability that each bidder wins the auction can be useful information for many people [7]. Algorithms have an understanding as a set of actions to solve a problem. The application of Algorithms is usually done in Machine Learning (ML) processes, which are related to systems that can provide performance from experience [8]. Machine learning (ML) is a type of artificial intelligence (AI) technology that allows a system to collect information without special programming [9]. The purpose of ML is two-fold: learn about different learning methods (eg the human method); and assist computers with similar functions [10]. The distinctive feature of Machine Learning itself is that there called training or learning process [10][11].

Problems with predicting structured data in the real world really require machine learning to process and validate data. However, these predictions are often constrained by limited real-world implementation due to a lack of tools to bridge the machine learning process [12]. Machine Learning requires

experience data to be studied as training data, in this case, namely experience data from participating in project tenders participated in by PT Rekayasa Industri.

The Random Forest classification is a development from CART method, namely by applying bootstrap aggregating (bagging) and random feature selection methods. In a random forest, many trees are grown to form a forest, then an analysis is performed on this collection of trees [13]. Logistic Regression is a predictive statistical technique, used to predict the relationship between dependent variables and independent variables, and Decision Tree is a prediction method that uses a tree model and possible consequences including accidental event outcomes, resource costs, and utilities.

The Logistic Regression, Decision Tree, and Random Forest algorithms were chosen as text classification algorithms compared to those not selected by the author because the accuracy of these three algorithms is relatively good based on [14] and some similar cases (based on literature studies) can be solved by these three algorithms.

II. METHODS

This framework suggested by this study was used to build the predictive models. It consists of three different stages: Data collection stage (Stage I), Data pre-processing stage (Stage II), and the predictive model development stage (Stage III). This means that the dataset for the model development goes through three different stages, Stage I, Stage II, and Stage III [10][15]. This research performs several stages in developing the design.

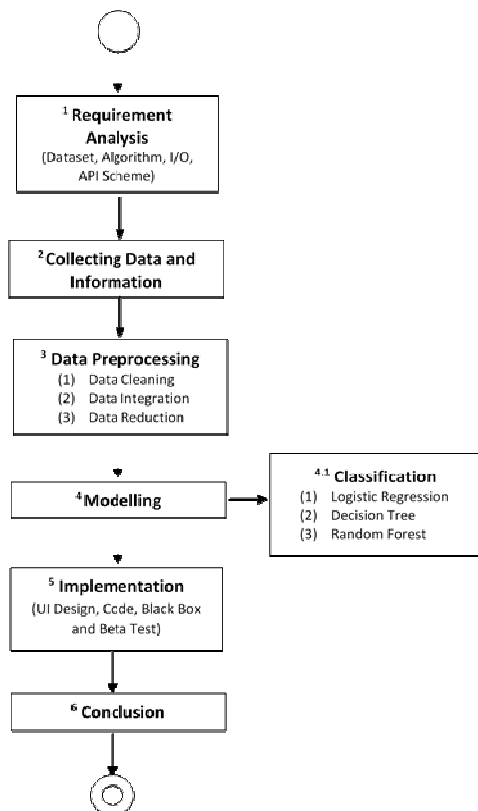


Fig 1. Methodology

Fig 1 Describes the early stages of doing a requirements analysis. Collecting data and information. The next stage is data preprocessing. The resulting data set is modeled using a classification algorithm[16]. The implementation stage is

doing the user interface design and writing the program. After that, test the system using blackbox testing and beta testing.

The proposed model in this study uses a combination of data mining and machine learning algorithms [17]. As shown in the flow diagram in figure 1.

• Requirement Analysis

Prepare tender input data from dataset. This study created 100 dummy data based on the pattern of the 30 original data provided in order to increase the predictive ability of the model to be used. So that we get 130 data. this study uses the ready-to-use Scikit-learn library of machine learning models in order to be able to present, operate, and provide output from the Logistic Regression, Decision Tree, and Random Forest models.

• Collecting Data and Information

Experience data recorded in the excel file from the company. data already has variables that will be used to predict. such as price, partners and competitors. Price is the value of the tender project implemented. then Partner, is participation and tender implementation. and finally competitors, is the condition of companies that are opponents of PT Rekayasa Industri and partners.

TABLE I. DATASET

Dataset	
Structure	Isi structure
Price	Low = 0
Partner	Moderate = 1
Competitor	Optimistic = 2
Result	Win or Lose

• Data Preprocessing

Data pre-processing of the tender dataset is necessary due to the fact that information has not been verified automatically to correct human errors, such as incorrect formatting, wrong values, empty fields, and so on [18]. data pre-processing stage, 130 experience data will be manipulated on the text that goes into the machine learning process, including data cleaning, case folding, and labelling processes.

TABLE II. DATA PRE-PROCESSING

#	Price	Partner	Competitor	Result
1	low	low	low	win
2	low	low	moderate	win
3	low	moderate	optimist	lose
...				
128	low	low	low	win
129	low	optimist	low	win
130	moderate	optimist	optimist	lose

• Modelling

To be able to compare which algorithm can give the best score in terms of studying the data. The following is the process of using machine learning by users. From all the pre-processing stages, the data is then used to train the model that will be used, namely

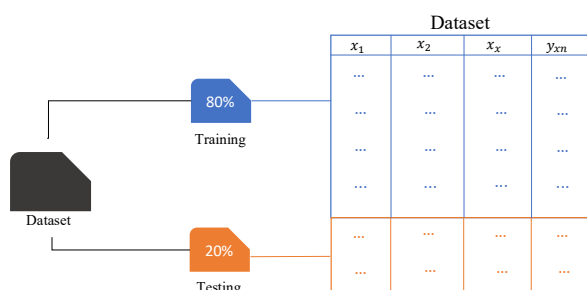
The diagram shows a central blue vertical bar labeled "Client Service". To its left is a box containing the text "I want to use **LR**, and **RF** to ensure bidding on future projects" and a smiley face icon. A blue arrow points from the box to the "Client Service" bar, and another blue arrow points from the bar back to the box. To the right of the "Client Service" bar are three model components, each consisting of a chip icon and text:

- LR**: Status: **Processing** (green text). A blue arrow points from the "Client Service" bar to this component, and another blue arrow points back.
- DT**: Status: **Idle** (blue text). No arrows are shown between the "Client Service" bar and this component.
- RF**: Status: **Processing** (green text). A blue arrow points from the "Client Service" bar to this component, and another blue arrow points back.

III. RESULT

A. Collecting Data

The data collected is divided into two parts of data (splitting data) namely training data (training data) and test data (testing data) with a portion of 80 to 20 (80:20). So that there are 26 experience data for testing purposes and 104 data for training purposes, each of which is divided into 4 (four) classes (data structure).



B. Data Preprocessing

- **Cleaning data [21].** It is necessary to clean the data on the dataset before it is processed by preparing data that is still in an irregular state and discarding data characters. This is done to avoid potential errors generated by the model.
- **Labelling Data.** It is necessary to label various data so that the machine learning model can properly recognize the data entry by assigning a meaningful identity (label) to the data

Data -	Price	Partner	Competitor	Result
1	0	0	0	1
2	0	0	1	1
3	0	0	2	0

. . . .				
128	0	2	1	1
129	0	2	0	1
130	1	2	2	0

- Logistic Regression

The figure consists of two side-by-side line plots. The left plot, titled 'Win Prediction Probability', shows a blue line representing the win probability over 25 iterations. The y-axis is labeled 'result' and ranges from 0.0 to 1.0. The line starts at 1.0, drops sharply to about 0.2 at iteration 2, then fluctuates between 0.2 and 1.0, ending at approximately 0.95. The right plot, titled 'Lose Prediction Probability', shows a red line representing the lose probability over 25 iterations. The y-axis is labeled 'result' and ranges from 0.0 to 0.8. The line starts at 0.0, peaks at about 0.85 at iteration 2, then fluctuates between 0.0 and 0.85, ending at approximately 0.95.

Fig 4 shows prediction result using test data, produce modeling uses 25 tender participation experience data for further predictions. The goal is to find out the prediction results using test data to support the model development process as well as material for observing what will happen to the model.

- One of the benefits of using a Decision Tree classification [23] is that the output is easy to understand and visualize with the library.

the condition of competitors will greatly affect the prediction results later with a value of 0.195. Prices with an interest value of 0.071, while the lowest is the partner factor with an importance value of 0.130. This figure can provide insight that competitors greatly influence the opportunities that a company can win in a project tender.

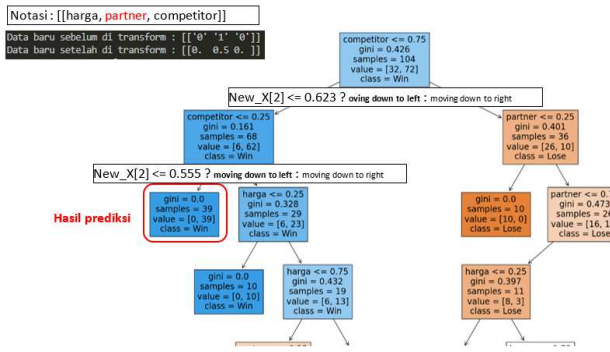


Fig 6. Visualization of the Decision Tree prediction process

- Random Forest

Random Forest [23][13] can be able to provide more optimal prediction results with a model whose parameters have been configured by displaying different feature importance values.

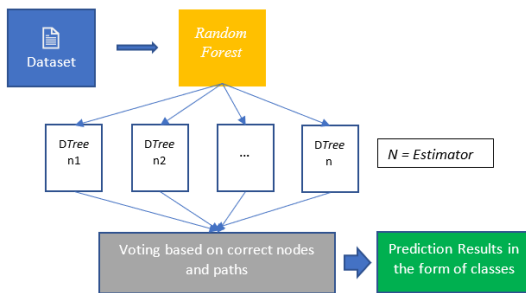


Fig 7. Illustration of Random Forest Prediction Flow

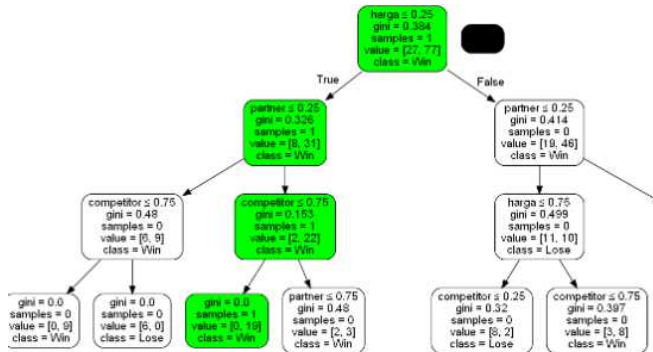


Fig 8. Visualization of Path Tree from Random Forest

Fig 7 show the first Decision Tree path of the Random Forest based on the new state input displays the predicted winning class.

D. Implementation

- User Interface

The real application display when the user sees the predicted results with an algorithm that follows the display that was previously designed at the interface design stage.

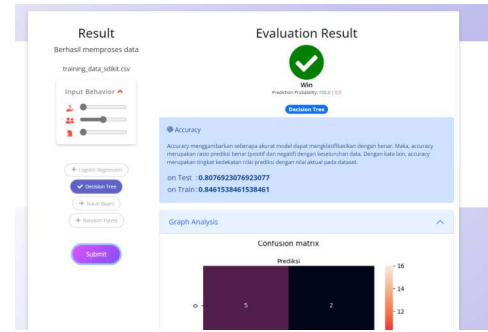


Fig. 8 Output Predictive Data and Information

- API Response

The API Response stage is to ensure that the dataset and input go through the API scheme with the POST method to the /predict function and the input can be detected. The data properties received by the /predict function are checkbox algorithms (in the form of plain text with a comma symbol as a separator for the model chosen by the user), price (in the form of integers), partners (in the form of integers), competitors (in the form of integers), and input_file (in the form of files).

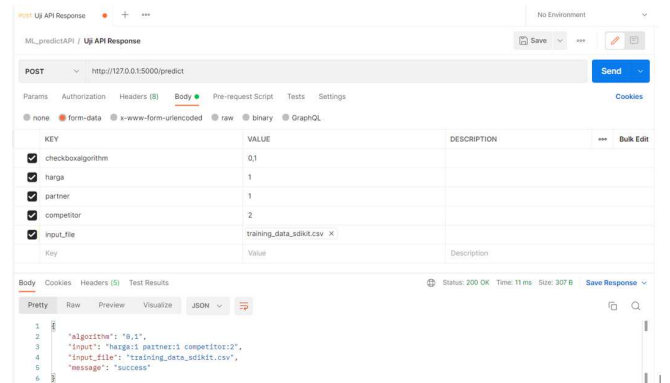


Fig 9. API Testing

IV. CONCLUSIONS

Based on a predetermined plan, the design implements a separate service system by separating applications (front-end) and machine learning services (back-end) where data flowing between the front end and back end must go through the API communication scheme, starting from defining input actions on the front-end side, namely the state level input (totaling three, namely price, partners, competitors) and an excel file in csv format. The input data flows through the API communication scheme and then is processed on the back-end side to carry out actions such as data validation, preprocessing, model training, and returning prediction results through the API communication scheme again until the application receives the data and can be seen by system users. Authors and Affiliations.

From the results of the tests carried out, Random Forest is a suitable model in this case, namely predicting victory, being able to study experience data well, and of course being a model that can help minimize the analysis process carried out by the marketing department.

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